

Layer Selection, Adaptation, and Discretization of Self-Supervised Speech Representations for Low-Resource ASR

Jiamin Qiu (Student ID: 524030910068)¹

¹*Shanghai Jiao Tong University, Shanghai, China*
qiujm_2024@sjtu.edu.cn

Keywords: Low-Resource ASR, Self-Supervised Speech Representation, Wav2Vec2, WavLM, CTC, Frozen Representation, Discrete Speech Units, Bitrate.

Abstract: This paper studies how self-supervised speech representations should be adapted, selected, and discretized for low-resource automatic speech recognition. We build a character-level CTC system with pretrained Wav2Vec2 and WavLM encoders and evaluate it on LibriSpeech subsets with 1.001, 3.001, and 6.392 effective hours of labeled speech. Beyond the main fine-tuning baseline, we compare partial encoder adaptation, frozen layer-wise probing, downstream head capacity, learned layer mixtures, and K-means discrete units. Wav2Vec2 full fine-tuning reaches 11.44% WER on test-clean, while masking gives a statistically comparable 11.35%. Updating the top 5–8 Transformer layers preserves most of the full fine-tuning performance, and frozen layer-8 features with a BiLSTM head remain effective with only 3.695M trainable parameters. Discrete units reduce representation bitrate by orders of magnitude but still lag continuous middle-layer features.

1 INTRODUCTION

Automatic speech recognition (ASR) systems typically require substantial paired speech–text data. This requirement is restrictive in low-resource settings, where limited supervision reduces acoustic, phonetic, speaker, and lexical coverage. Self-supervised speech encoders provide a practical alternative: they learn reusable frame-level representations from unlabeled audio and can be adapted to ASR with much less labeled data. Wav2Vec2 is a representative example, showing that pretraining on raw speech followed by limited supervised fine-tuning is effective for low-resource recognition (Baevski et al., 2020b).

Using a pretrained speech encoder is not a single design choice. The encoder may be fully fine-tuned, only partially updated, or kept frozen while a lightweight downstream model is trained. If the encoder is frozen, the selected hidden layer and the capacity of the downstream head can dominate recognition performance. In addition, continuous hidden states can be quantized into discrete units, which may reduce representation bitrate but can also remove information needed by CTC alignment and lexical discrimination.

This paper studies these choices in a controlled low-resource English ASR setting. We use LibriSpeech (Panayotov et al., 2015), pretrained

Wav2Vec2 and WavLM encoders (Baevski et al., 2020b; Chen et al., 2022), and character-level CTC decoding (Graves et al., 2006). The experiments compare full fine-tuning, top- k partial fine-tuning, frozen layer-wise probing, downstream head variants, learned layer mixtures, and K-means discrete units. The goal is not to propose a new ASR architecture, but to identify how self-supervised representations should be used when labeled speech is scarce.

The main findings are as follows. First, Wav2Vec2 full fine-tuning gives the strongest 6.392-hour ASR performance, while partial adaptation of the top 5–8 Transformer layers recovers most of that accuracy with fewer trainable parameters. Second, frozen representations are useful only when both layer and head are chosen carefully: middle layers outperform the final layer, and a temporal BiLSTM head is much stronger than a linear probe. Third, K-means discrete units form a clear bitrate–WER trade-off; larger codebooks improve recognition, but even 1000-unit centroids remain worse than matched continuous representations.

2 RELATED WORK

2.1 Self-Supervised Speech Representation Learning

Self-supervised speech models pretrain encoders on unlabeled waveforms and transfer the learned representations to downstream speech tasks. Wav2Vec2 masks latent speech representations and solves a contrastive prediction task over quantized targets (Baevski et al., 2020b). HuBERT instead uses off-line clustering to generate masked prediction targets and relies on cluster consistency rather than perfect unit labels (Hsu et al., 2021). WavLM extends masked speech prediction with denoising and gated relative position bias to support a wider range of speech-processing tasks (Chen et al., 2022). Benchmarks such as SUPERB evaluate the generality of frozen SSL representations with lightweight task-specific heads (Yang et al., 2021). Our work follows this representation-analysis view, but focuses on low-resource ASR and explicitly compares layer choice, head capacity, and adaptation scope.

2.2 Low-Resource ASR and CTC Fine-Tuning

Connectionist temporal classification (CTC) trains sequence recognizers without frame-level alignments by marginalizing over all valid input-output alignments (Graves et al., 2006). This makes it a simple and common objective for adapting SSL encoders to ASR. In low-resource settings, however, full fine-tuning can be sensitive to optimization and data scale, while frozen features reduce the number of updated parameters but may require a stronger downstream model. This paper therefore treats full fine-tuning, partial fine-tuning, and frozen probing as distinct adaptation regimes rather than as a single baseline comparison.

2.3 Discrete Speech Units and Representation Compression

Discrete speech units convert continuous acoustic representations into symbolic or prototype-based sequences. vq-wav2vec showed that dense speech representations can be quantized with Gumbel-softmax or online K-means, enabling discrete-input modeling methods (Baevski et al., 2020a). Generative spoken language modeling further demonstrated that pseudo-text units extracted from speech can support textless modeling pipelines, and that the number of discrete

units affects downstream behavior (Lakhotia et al., 2021). In this work, discrete units are used for ASR representation analysis rather than speech generation. We measure their token rate and bitrate, compare multiple codebook sizes, and distinguish centroid inputs from learned unit embeddings.

3 PROBLEM FORMULATION AND SYSTEM OVERVIEW

3.1 ASR Task Definition

Given an input waveform $x = (x_1, x_2, \dots, x_T)$, the ASR system predicts a normalized text sequence $y = (y_1, y_2, \dots, y_N)$. In this work, y is represented as a character sequence over an English CTC vocabulary.

3.2 CTC Recognition Objective

The model estimates the posterior probability $P(y|x)$ using frame-level CTC logits. The CTC loss marginalizes over all valid frame-level alignments π that collapse to the target sequence y :

$$\mathcal{L}_{\text{CTC}} = -\log \sum_{\pi \in \mathcal{B}^{-1}(y)} \prod_t P(\pi_t | x), \quad (1)$$

where $\mathcal{B}$ denotes the CTC collapse operation that removes repeated symbols and blanks.

3.3 Overall SSL-ASR Pipeline

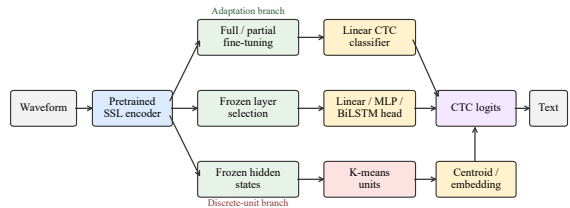


Figure 1: Overview of the SSL-based low-resource ASR system.

The overall system first extracts speech representations using a pretrained SSL encoder. Depending on the experimental condition, the representation is either fine-tuned end-to-end, used as a frozen continuous hidden state, or quantized into discrete units. A CTC-based downstream model then predicts the final transcription.

4 METHODOLOGY

4.1 SSL Encoder with CTC Head

Let f_θ denote a pretrained Wav2Vec2 or WavLM encoder. It consists of a convolutional feature extractor, a feature projection, and 12 Transformer blocks. For an input waveform x , it produces a sequence of 768-dimensional hidden states

$$H^{(L)} = f_\theta(x) = (h_1^{(L)}, \dots, h_{T'}^{(L)}), \quad (2)$$

where $L = 12$ for the final Transformer block and $T' \ll T$ because of convolutional subsampling. A linear classifier maps each frame to the CTC label space $\mathcal{V}' = \mathcal{V} \cup \{\emptyset\}$:

$$p_t = \text{softmax}(Wh_t^{(L)} + b). \quad (3)$$

Here, $\mathcal{V}$ contains lowercase English characters, apostrophe, and the word-boundary symbol, and $\emptyset$ is the CTC blank. The encoder and classifier are optimized with Eq. 1. At inference, frame-wise argmax predictions are collapsed by $\mathcal{B}$; no lexicon or external language model is used.

4.2 Full, Partial, and Frozen Adaptation

We vary the trainable subset of the same pretrained encoder to separate representation reuse from encoder adaptation. The convolutional feature extractor is fixed in all conditions.

Full Fine-Tuning. All 12 Transformer blocks, the feature projection, and the randomly initialized CTC classifier are updated. Thus, “full” refers to the contextual encoder rather than the convolutional front end.

Partial Fine-Tuning. For top- k adaptation, the bottom $12 - k$ Transformer blocks are fixed, while the feature projection, top k blocks, and CTC classifier are updated. Sweeping k produces a controlled accuracy–trainable-parameter curve without changing encoder depth at inference.

Frozen Representation Probing. The complete SSL encoder is fixed and only a downstream CTC head is trained. Hidden states are cached once for head training, which prevents encoder dropout or gradient flow from confounding the comparison. At test time, however, the encoder is still executed. Frozen probing therefore reduces the number of trainable parameters, but does not by itself reduce the deployed encoder size.

4.3 Layer-Wise Representation Probing

We denote the projected convolutional features by $H^{(0)}$ and the output of Transformer block l by $H^{(l)}$, $l \in \{1, \dots, 12\}$. For each candidate layer, the encoder is frozen and an independently initialized head of identical architecture is trained:

$$P(y | x; l) = g_{\phi_l}(H^{(l)}). \quad (4)$$

Layer ranking and all subsequent design choices use development WER only; test results do not participate in model selection. Keeping the head fixed attributes differences to layer content rather than downstream capacity.

Learned Layer Mixture. As an alternative to selecting one layer, we learn a global convex combination over all 13 representations:

$$\tilde{h}_t = \sum_{l=0}^{12} a_l h_t^{(l)}, \quad a_l = \frac{\exp(\alpha_l)}{\sum_{j=0}^{12} \exp(\alpha_j)}, \quad (5)$$

where the scalar logits α_l are initialized uniformly and learned jointly with the downstream head. The SSL hidden states remain fixed.

4.4 Downstream Head Variants

To distinguish representation quality from decoder capacity, we compare four CTC heads on the same frozen layer. The linear head applies a single affine projection. The MLP head applies

$$o_t = W_2 \text{GELU}(W_1 h_t + b_1) + b_2. \quad (6)$$

It adds frame-wise nonlinearity but no temporal context. The recurrent variants use either one or two bi-directional LSTM layers with 256 hidden units per direction, followed by a linear projection:

$$o_{1:T'} = W_o \text{BiLSTM}(H^{(l)}) + b_o. \quad (7)$$

Dropout is applied before the head and before the output projection; padded frames are excluded through packed variable-length sequences. All variants preserve the encoder frame rate and are trained with the same CTC targets.

4.5 K-Means Discrete Unit Modeling

We quantize hidden states from the layer selected by frozen probing. A MiniBatch K-means codebook $C = \{c_1, \dots, c_K\}$ minimizes

$$\min_C \sum_{h \in S_{\text{train}}} \min_{k \in \{1, \dots, K\}} \|h - c_k\|_2^2, \quad (8)$$

where $\mathcal{S}_{\text{train}}$ contains at most 500,000 frames sampled exclusively from the training split. Development and test frames never contribute to codebook fitting. Each frame is then replaced by its nearest-centroid index

$$z_t = \arg \min_k \|h_t - c_k\|_2^2. \quad (9)$$

The raw frame-rate unit sequence is retained for ASR; consecutive duplicates are removed only when reporting the optional deduplicated bitrate.

Centroid Input. The centroid condition reconstructs each frame as

$$\tilde{h}_t = c_{z_t}. \quad (10)$$

It preserves the codebook geometry while restricting the representation to K prototypes.

Learned Unit Embedding. The embedding condition instead treats z_t as a categorical symbol:

$$\tilde{h}_t = E[z_t], \quad (11)$$

where E is learned jointly with the same downstream CTC head. Comparing these inputs at fixed layer, K , vocabulary, and training schedule tests whether centroid geometry carries information beyond unit identity.

4.6 Bitrate and Complexity Measurement

Let N be the number of unit frames in D seconds and $\hat{p}_k$ the empirical frequency of unit k . We report unit rate $r = N/D$, entropy

$$H(U) = - \sum_{k=1}^K \hat{p}_k \log_2 \hat{p}_k, \quad R_{\text{unit}} = rH(U), \quad (12)$$

in bits per second. We additionally compute the fixed-width rate $r \lceil \log_2 K \rceil$ and repeat the entropy-rate calculation after run-length deduplication. For a d -dimensional continuous representation stored with b bits per component, the corresponding uncompressed rate is rdb . Codebook utilization is the fraction of centroids observed, and unit perplexity is $2^{H(U)}$.

Efficiency is characterized by both trainable and total parameter counts. The former measures adaptation cost, whereas the latter reflects the complete encoder-head system. End-to-end real-time factor is measured as synchronized inference wall time divided by total audio duration and includes SSL encoding, nearest-centroid assignment when applicable, and CTC-head inference. The reported unit bitrate describes only the intermediate representation’s theoretical storage or transmission rate; it is not a measure of model size or inference cost.

5 EXPERIMENTAL SETUP

5.1 Dataset and Preprocessing

We use the 16-kHz LibriSpeech corpus for low-resource English ASR. Nested training subsets are sampled from `train-clean-100` with seed 42. Utterances longer than 15 s are excluded before duration accumulation; consequently, the nominal 10-hour manifest contains 6.392 effective hours after filtering. The same duration limit is applied to `dev-clean` during training, whereas the complete `test-clean` split is retained for final evaluation.

Table 1: Dataset statistics used in the experiments.

Split	Usage	Utterances	Hours
Train-1h	Training	315	1.001
Train-3h	Training	953	3.001
Train-10h	Training	2012	6.392
Dev-clean (≤ 15 s)	Selection	2513	4.364
Test-clean	Test	2620	5.403

Waveforms are converted to mono when necessary and normalized by the pretrained model’s feature extractor. Transcripts are lowercased; punctuation other than apostrophes is removed and repeated whitespace is collapsed. The fixed 30-symbol CTC vocabulary comprises 26 letters, apostrophe, a word-boundary symbol, an unknown symbol, and the blank/padding symbol. No data augmentation other than the explicitly studied time- and feature-masking conditions is applied.

5.2 Model Configurations

The primary encoder is `facebook/wav2vec2-base`; `microsoft/wavlm-base` provides an encoder comparison. Both are self-supervised base checkpoints rather than supervised ASR checkpoints and contain 12 Transformer blocks with 768-dimensional states. Full and partial fine-tuning use a linear CTC classifier. Frozen continuous and discrete experiments use cached Wav2Vec2 states and, unless head capacity is the variable under study, a two-layer BiLSTM with 256 units per direction and dropout 0.1. K-means codebooks contain $K \in \{50, 100, 200, 500, 1000\}$ units; learned unit embeddings use 256 dimensions, with a 768-dimensional control. Table 2 summarizes the experimental groups.

5.3 Training Details

All systems use AdamW, mean CTC loss with infinite losses mapped to zero, linear warm-up and decay, and

Table 2: Experimental groups and corresponding research questions.

Group	Research Question	Experiments
Main ASR	Does SSL fine-tuning work for low-resource ASR?	E1, E2, E3, E4, E5, E8
Data scale	How does labeled data scale affect WER?	E3, E5, E2
Adaptation	Is full fine-tuning necessary?	E2, E6A, E6B, E16A–C, E25
Layer probing	Which hidden layer is best for frozen ASR?	E10, E13, E14, E23
Head capacity	How strong should the downstream head be?	E15
Discrete units	What is the WER–bitrate trade-off?	E11, E17, E18, E24
Error analysis	Which error types dominate?	E21
Stability	How stable are key findings?	E22, E23, E24, E25

gradient clipping at 1.0. Fine-tuning uses mixed precision and Transformer gradient checkpointing. The 6.392-hour Wav2Vec2/WavLM runs use learning rate 10^{-4} , weight decay 0.005, warm-up ratio 0.1, effective batch size 16, and 30 epochs. The 3-hour run uses 2000 updates, learning rate 7×10^{-5} , and warm-up ratio 0.15. To stabilize the 1-hour run, the CTC head and encoder use learning rates 10^{-4} and 2×10^{-5} , respectively; the encoder is held fixed for the first 100 of 1500 updates. Partial fine-tuning uses 2500 updates with learning rate 10^{-4} .

Cached-representation heads are trained for at most 3000 updates with batch size 16, learning rate 10^{-3} , weight decay 0.005, and warm-up ratio 0.1; the one-layer BiLSTM uses 8×10^{-4} . Codebooks are fitted with three MiniBatch K-means initializations on at most 500,000 training frames. Seed 42 is used throughout, with seed 1234 reruns for the selected frozen layer and top-5 partial adaptation. Training was conducted on CUDA GPUs, primarily RTX 3090/3090 Ti devices.

Development WER determines layer, head, codebook, and adaptation choices. Step-based experiments retain the checkpoint with the lowest development WER; fixed 30-epoch encoder comparisons use their final checkpoint. The untouched test split is evaluated only after each configuration is fixed, using greedy CTC decoding without an external language model.

5.4 Evaluation Metrics

Word error rate, $WER = (S + D + I)/N$, is the primary metric; character error rate (CER) and exact-match utterances are secondary metrics. We also report substitution, deletion, and insertion counts, trainable and total parameters, end-to-end RTF, and the representation statistics defined in Section 4. Selected system differences are assessed with 10,000-sample paired utterance bootstrap 95% confidence intervals. All reported text metrics are computed after applying the same normalization to hypotheses and references.

6 RESULTS AND ANALYSIS

6.1 Main ASR Results

Table 3 establishes full Wav2Vec2 fine-tuning as the strongest main configuration. E8 has the lowest point estimate, but its -0.09 percentage-point (pp) difference from E2 has a paired-bootstrap 95% confidence interval (CI) of $[-0.31, +0.14]$ pp. We therefore regard E2 and E8 as equivalent rather than attributing a reliable gain to masking. The same conclusion holds at 1 hour: E12A differs from E3 by only $+0.23$ pp, with CI $[-0.05, +0.50]$ pp.

Under the matched 6.392-hour setting, WavLM reaches 17.50% WER versus 11.44% for Wav2Vec2. This is a configuration-specific result: it shows that replacing the pretrained encoder without retuning does not guarantee improvement, not that Wav2Vec2 is universally superior. Conversely, E1 shows that the final frozen layer and a linear head are inadequate; later controlled probes show that frozen representations themselves remain useful.

Table 3: Main ASR results on LibriSpeech test-clean. Parameters are the number updated during training.

ID	System	Hours	Trainable Params	WER (%)	CER (%)
E1	Frozen Wav2Vec2 + linear CTC	6.392	0.023M	99.92	85.46
E3	Wav2Vec2 full fine-tuning	1.001	90.194M	40.51	11.65
E12A	Wav2Vec2 full FT + time masking	1.001	90.194M	40.73	11.67
E5	Wav2Vec2 full fine-tuning	3.001	90.194M	17.12	4.88
E2	Wav2Vec2 full fine-tuning	6.392	90.194M	11.44	3.36
E8	Wav2Vec2 full FT + masking	6.392	90.194M	11.35	3.34
E4	WavLM full fine-tuning	6.392	90.205M	17.50	5.08

6.2 Data Scale and Adaptation Regime

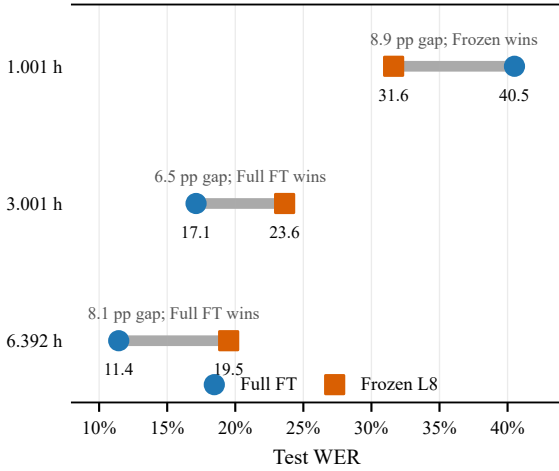


Figure 2: Interaction between labeled-data scale and adaptation strategy. Horizontal spans show the absolute WER gap and the preferred strategy at each scale.

Figure 2 combines the data-scale and adaptation results. For full fine-tuning, increasing labeled speech from 1.001 to 3.001 hours reduces WER from 40.51% to 17.12%, a 57.7% relative reduction. Increasing it further to 6.392 hours yields 11.44%, an additional 33.2% relative reduction. The first few labeled hours are thus most valuable, but performance has not saturated.

The preferred adaptation strategy changes with data scale. At 1 hour, frozen layer 8 with a BiLSTM obtains 31.63% WER, outperforming full fine-tuning by 8.88 pp (95% CI $[-9.45, -8.29]$ pp for frozen minus full). At 3.001 and 6.392 hours, full fine-tuning is better by 6.50 and 8.08 pp, respectively. Updating approximately 90M parameters is difficult to regularize with only one hour of labels; once supervision increases, task-specific encoder adaptation outweighs that disadvantage.

6.3 Trainable Parameter Budget

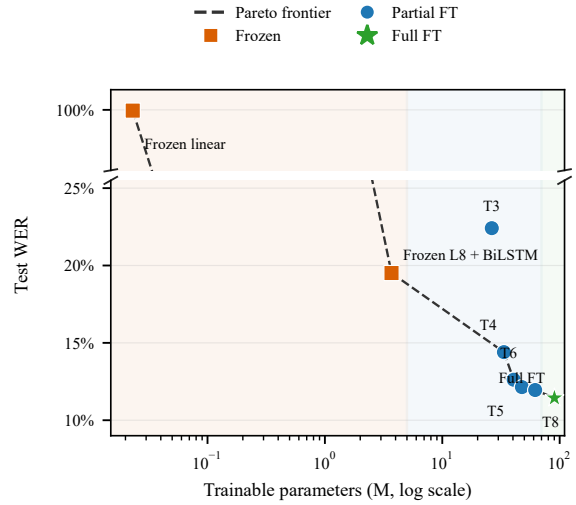


Figure 3: Parameter efficiency across frozen, partial, and full adaptation. The logarithmic axis exposes the low-parameter regime; the dashed line marks the empirical Pareto frontier.

Figure 3 shows a nonlinear return from unfreezing Transformer blocks. Top-3 adaptation uses 26.403M trainable parameters but reaches only 22.41% WER. Unfreezing one more block lowers WER sharply to 14.40%; top-5, top-6, and top-8 then reach 12.62%, 12.14%, and 11.95%, respectively. Top-8 is closest to full fine-tuning, while top-5 provides a stronger parameter-accuracy compromise, using 45% of the full trainable parameters for a 1.18 pp WER cost. The seed-1234 top-5 rerun gives 13.02%, compared with 12.62% at seed 42, indicating modest seed sensitivity without changing the ranking.

Frozen layer 8 occupies a distinct low-update regime: its BiLSTM head trains only 3.695M parameters and reaches 19.51% WER. This does not imply a smaller inference model. Partial systems execute every Transformer block, and the frozen BiLSTM system has 98.066M total parameters versus 94.395M for

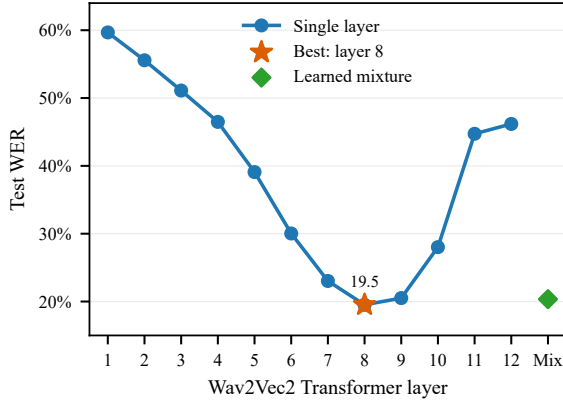


Figure 4: Layer-wise WER of frozen Wav2Vec2 representations with the same BiLSTM-CTC head.

the linear full-fine-tuning system.

6.4 Layer-Wise Analysis of SSL Representations

The probing curve in Figure 4 is strongly layer-dependent. WER decreases from 59.67% at layer 1 to 19.51% at layer 8, then rises to 46.18% at layer 12. Layer 9 remains competitive at 20.51%, defining a middle-layer “sweet spot.” The seed-1234 layer-8 rerun reaches 19.59%, only 0.08 pp above seed 42. Thus, the optimum is not an isolated initialization artifact.

The learned mixture reaches 20.35% WER and assigns its largest weights to layers 7–9, consistent with the probe. It nevertheless trails hard layer-8 selection by 0.83 pp, suggesting that the small labeled set cannot fully suppress weaker layers. These results also explain why a frozen final-layer baseline understates representation quality: later layers are specialized toward the pretraining objective and are not necessarily the most linearly accessible for ASR.

6.5 Effect of Downstream Head Capacity

Table 4: Effect of downstream head capacity on frozen layer-8 representations.

Head	Trainable Params	WER (%)	CER (%)
Linear CTC	0.025M	65.12	23.12
MLP CTC	0.410M	38.15	10.99
1-layer BiLSTM	2.118M	23.98	7.42
2-layer BiLSTM	3.695M	19.57	6.13

Table 4 separates frozen-feature quality from head capacity. Adding frame-wise nonlinearity reduces

WER from 65.12% to 38.15%, while adding bidirectional temporal context produces a further large reduction. The two-layer BiLSTM reaches 19.57%, essentially reproducing the independent layer-8 probe. Therefore, E1 fails because it combines a weak linear head with an unsuitable final layer, not because frozen Wav2Vec2 contains no ASR-relevant information.

6.6 Continuous versus Discrete Speech Representations

Figure 5 shows a consistent accuracy–rate trade-off: WER falls from 71.56% at $K = 50$ to 33.08% at $K = 1000$ as empirical bitrate increases from approximately 275 to 462 bit/s. The strict layer-8 comparison confirms that increasing K from 200 to 1000 improves WER by 14.36 pp (95% CI $[-14.82, -13.91]$ pp for $K = 1000$ minus $K = 200$). Nevertheless, $K = 1000$ remains 13.40 pp worse than the matched continuous representation. High codebook utilization (92.2% at $K = 1000$) indicates that this gap is not caused simply by unused clusters.

At fixed layer 8 and $K = 200$, the centroid vector outperforms the 256-dimensional learned embedding by 2.38 pp (95% CI $[2.01, 2.75]$ pp). Increasing the embedding to 768 dimensions worsens WER by another 1.45 pp. Retaining centroid geometry is therefore more useful than treating unit IDs as unrelated symbols under this training budget. The unit stream is over 1300 times smaller in theoretical bitrate than an FP16 continuous state, but this is representation compression only: the encoder and nearest-centroid search are still required online.

Table 5: Controlled comparison of layer-8 continuous and discrete representations. Bitrate describes the unit-index stream; “dedup” removes consecutive repeats.

Input to CTC head	K	Input Dim.	Params	WER (%)	Bitrate	Dedup
Continuous state	–	768	3.694M	19.67	613,344	–
Centroid	200	768	3.694M	47.43	364.06	208.00
Centroid	1000	768	3.694M	33.08	461.93	286.57
Learned embedding	200	256	2.696M	49.81	364.06	208.00
Learned embedding	200	768	3.847M	51.26	364.06	208.00

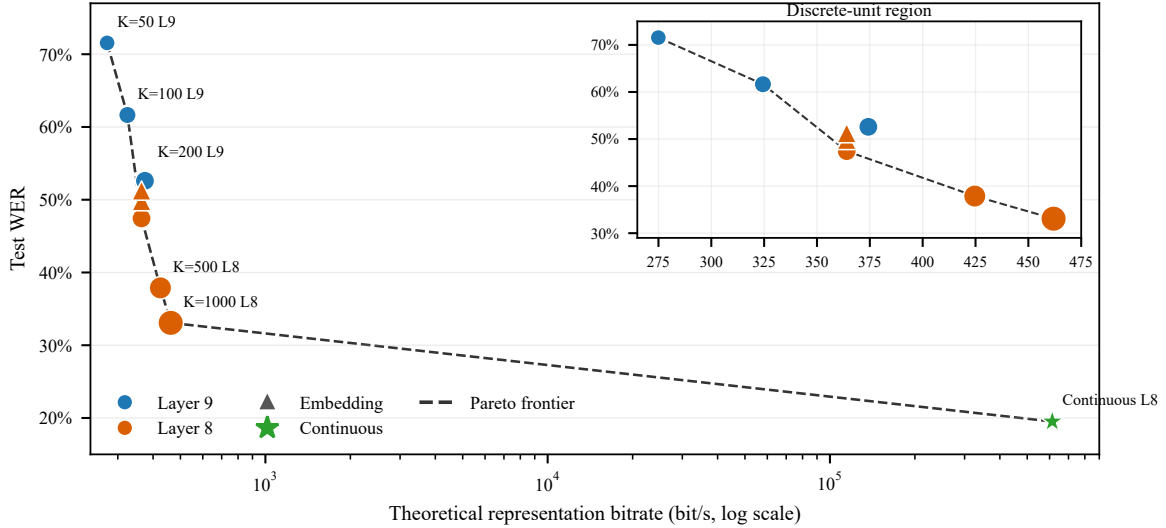


Figure 5: Rate–distortion frontier for continuous and discrete representations. The inset enlarges the discrete-unit region; circles denote centroid inputs and triangles denote learned embeddings. The $K \leq 200$ and $K \geq 500$ centroid points use layers 9 and 8, respectively.

6.7 Efficiency, Error Types, and Stability

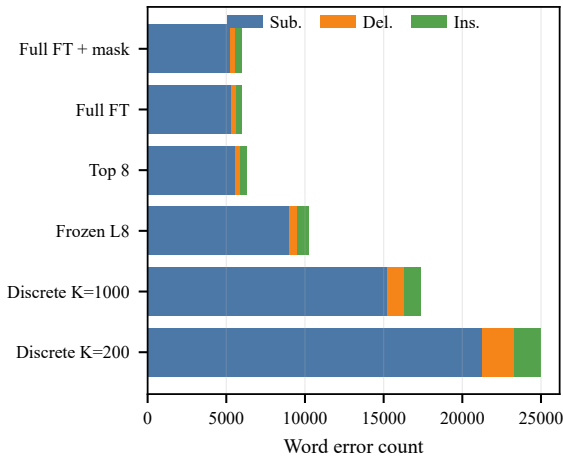


Figure 6: Word-level error composition for representative systems.

Across a common remote evaluation, linear-head Wav2Vec2 systems have RTF 0.00122–0.00125, whereas continuous and discrete BiLSTM pipelines have RTF 0.0036–0.0038. The close RTFs of continuous, $K = 200$, and $K = 1000$ systems show that K-means assignment adds little relative to SSL encoding and BiLSTM inference. Absolute RTFs measured on different GPUs are not directly comparable.

Figure 6 shows that substitutions dominate every effective system. As adaptation is restricted or the representation is quantized, deletions and insertions also increase: E8 records 311 deletions and 390 insertions, compared with 1,108 and 1,078 for $K = 1000$. Discretization therefore affects both lexical discrimination and CTC sequence alignment rather than producing only isolated near-word substitutions.

Finally, failed runs delimit the optimization claims. The online frozen BiLSTM run E9 developed non-finite gradients and collapsed, while two 1-hour masking-grid runs produced empty output. They are training failures, not valid upper bounds for their architectures. Conversely, the layer-8 result is stable across two seeds, while top-5 varies by 0.40

pp. Since most full-fine-tuning and discrete conditions have one seed, small point-estimate differences should not be interpreted as general seed-level significance; the paired bootstrap intervals quantify test-utterance sampling only.

7 CONCLUSION

This work examined how pretrained speech representations should be adapted, selected, and discretized for low-resource ASR. With 6.392 effective hours of labeled speech, full Wav2Vec2 fine-tuning achieved 11.44% WER, while masking produced a statistically indistinguishable 11.35%. Partial adaptation recovered most of this performance with fewer trainable parameters: top-5 fine-tuning reached 12.62% WER using 45% of the full update budget. The preferred strategy nevertheless depended on data scale; frozen layer 8 with a BiLSTM outperformed full fine-tuning at one hour, whereas full adaptation became substantially better from three hours onward.

The representation analyses show that encoder use is as important as encoder choice. Frozen ASR performance peaked at layer 8 rather than the final layer, and a sufficiently expressive temporal head reduced WER from 65.12% with a linear probe to 19.57% with a two-layer BiLSTM. K-means units reduced the theoretical intermediate bitrate by over three orders of magnitude, but even a 1000-unit codebook remained 13.40 percentage points behind the matched continuous representation. Centroid inputs consistently outperformed learned unit embeddings, indicating that preserving representation geometry matters under limited supervision.

Overall, SSL encoders offer several viable low-resource regimes rather than a single universally optimal configuration. Full fine-tuning is preferable when sufficient labels are available; frozen middle-layer features and partial adaptation provide practical alternatives under tighter data or training-memory constraints. These conclusions are limited by the single corpus, greedy decoding without a language model, and mostly single-seed full-fine-tuning and discrete experiments. Broader validation should therefore test additional datasets, encoders, seeds, and learned acoustic tokenizers.

REFERENCES

- Baevski, A., Schneider, S., and Auli, M. (2020a). vq-wav2vec: Self-supervised learning of discrete speech representations. In *International Conference on Learning Representations*.
- Baevski, A., Zhou, H., Mohamed, A., and Auli, M. (2020b). wav2vec 2.0: A framework for self-supervised learning of speech representations. In *Advances in Neural Information Processing Systems*, volume 33, pages 12449–12460.
- Chen, S., Wang, C., Chen, Z., Wu, Y., Liu, S., Chen, Z., Li, J., Kanda, N., Yoshioka, T., Xiao, X., Wu, J., Zhou, L., Ren, S., Qian, Y., Qian, Y., Zeng, M., Yu, X., and Wei, F. (2022). WavLM: Large-scale self-supervised pre-training for full stack speech processing. *IEEE Journal of Selected Topics in Signal Processing*, 16(6):1505–1518.
- Graves, A., Fernández, S., Gomez, F., and Schmidhuber, J. (2006). Connectionist temporal classification: Labelling unsegmented sequence data with recurrent neural networks. In *Proceedings of the 23rd International Conference on Machine Learning*, pages 369–376. ACM.
- Hsu, W.-N., Bolte, B., Tsai, Y.-H. H., Lakhota, K., Salakhutdinov, R., and Mohamed, A. (2021). HuBERT: Self-supervised speech representation learning by masked prediction of hidden units. *IEEE/ACM Transactions on Audio, Speech, and Language Processing*, 29:3451–3460.
- Lakhota, K., Kharitonov, E., Hsu, W.-N., Adi, Y., Polyak, A., Bolte, B., Nguyen, T.-A., Copet, J., Baevski, A., Mohamed, A., and Dupoux, E. (2021). On generative spoken language modeling from raw audio. *Transactions of the Association for Computational Linguistics*, 9:1336–1354.
- Panayotov, V., Chen, G., Povey, D., and Khudanpur, S. (2015). LibriSpeech: An ASR corpus based on public domain audio books. In *Proceedings of the 2015 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, pages 5206–5210. IEEE.
- Yang, S.-w., Chi, P.-H., Chuang, Y.-S., Lai, C.-I. J., Lakhota, K., Lin, Y. Y., Liu, A. T., Shi, J., Chang, X., Lin, G.-T., Huang, T.-H., Tseng, W.-C., Lee, K.-T., Liu, D.-R., Huang, Z., Dong, S., Li, S.-W., Watanabe, S., Mohamed, A., and Lee, H.-y. (2021). SUPERB: Speech processing universal performance benchmark. In *Proceedings of Interspeech 2021*, pages 3161–3165.